

E-Commerce Customer Intelligence & Revenue Analytics

Final Advanced Python Internship Project Report

Built from the supplied internship PDF and Excel dataset. Raw sheets were kept read-only.

Metric	Result
Raw Orders	20,025
Clean Orders	20,000
Duplicates Removed	25
Revenue	■713,218,937.20
Profit	■120,998,318.62
AOV	■35,660.95
Return Rate	4.86%
Cancellation Rate	4.92%

Cleaning & validation

Removed 25 exact duplicate rows. Missing discount_pct (100) was filled with median 15%; missing delivery_days (51) was filled with median 5 days. Independent formula checks match supplied calculations to rounding-level differences.

Key business findings

- Cleaned dataset contains 20,000 orders after removing 25 exact duplicates.
- Net revenue is ■713,218,937.20, profit is ■120,998,318.62, and average order value is ■35,660.95.
- Return rate is 4.86% and cancellation rate is 4.92%.
- 2025-05 is the highest-revenue month at ■64,197,771.49; 2025-02 is the lowest.
- 2025-03 has the strongest MoM revenue growth at 14.61%.
- Beauty leads category revenue at ■110,749,474.13.
- Electronics has the lowest category profit margin at 13.73%.
- Furniture has the highest category profit margin at 21.17%.
- Fashion has the highest return rate at 5.56%.
- Discount vs profit margin is negatively correlated ($r=-0.621$, $p<0.001$).
- Shipping-mode ANOVA is not significant ($p=0.4753$).
- Payment method vs return flag is not significant at 5% ($p=0.1233$).
- 92.59% of customers are repeat purchasers; repeat buyers average about 76.6 days between purchases.
- RFM segmentation produces five customer groups for differentiated retention actions.
- Return prediction is difficult because returns are a minority class and the tested models show weak discrimination.

Statistical tests

Test	Statistic	p-value	Conclusion
Discount vs Profit Margin	-0.6207	0.000000	Reject H0
AOV by Shipping Mode	0.7438	0.475320	Fail to reject H0
Payment Method vs Return	8.6627	0.123300	Fail to reject H0

10 actionable recommendations

- Control discounts on low-margin products using contribution-margin thresholds.
- Review high-revenue/low-margin products for pricing, sourcing and promotion changes.
- Use loyalty and cross-sell campaigns for Champions and Loyal Active customers.
- Trigger retention campaigns for At Risk Loyal and Hibernating customers.
- Plan campaigns before historically weak months and protect peak-month capacity.
- Investigate high-return categories for product quality and customer-expectation gaps.
- Do not assume faster shipping automatically increases order value; use evidence-based service design.
- Do not target return prevention using payment method alone.
- Add richer behavioral/product features before deploying return prediction.
- Create a monthly dashboard for revenue, profit, margin, returns, cancellations, RFM risk and ML metrics.

ML conclusion

Two classification models were compared with stratified 5-fold cross-validation and a held-out test set. Precision, recall, F1 and ROC-AUC are used because return is an imbalanced class. The tested models provide a baseline but are not strong enough for production deployment without richer features.